Toward verifiable postal voting

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Motivations

Why be interested in postal voting?

Medium stake ballots

Postal voting is widely used:

- Professional elections
- Trade union elections
- Associations
- University boards of directors
- Political primaries



... but also very high stakes

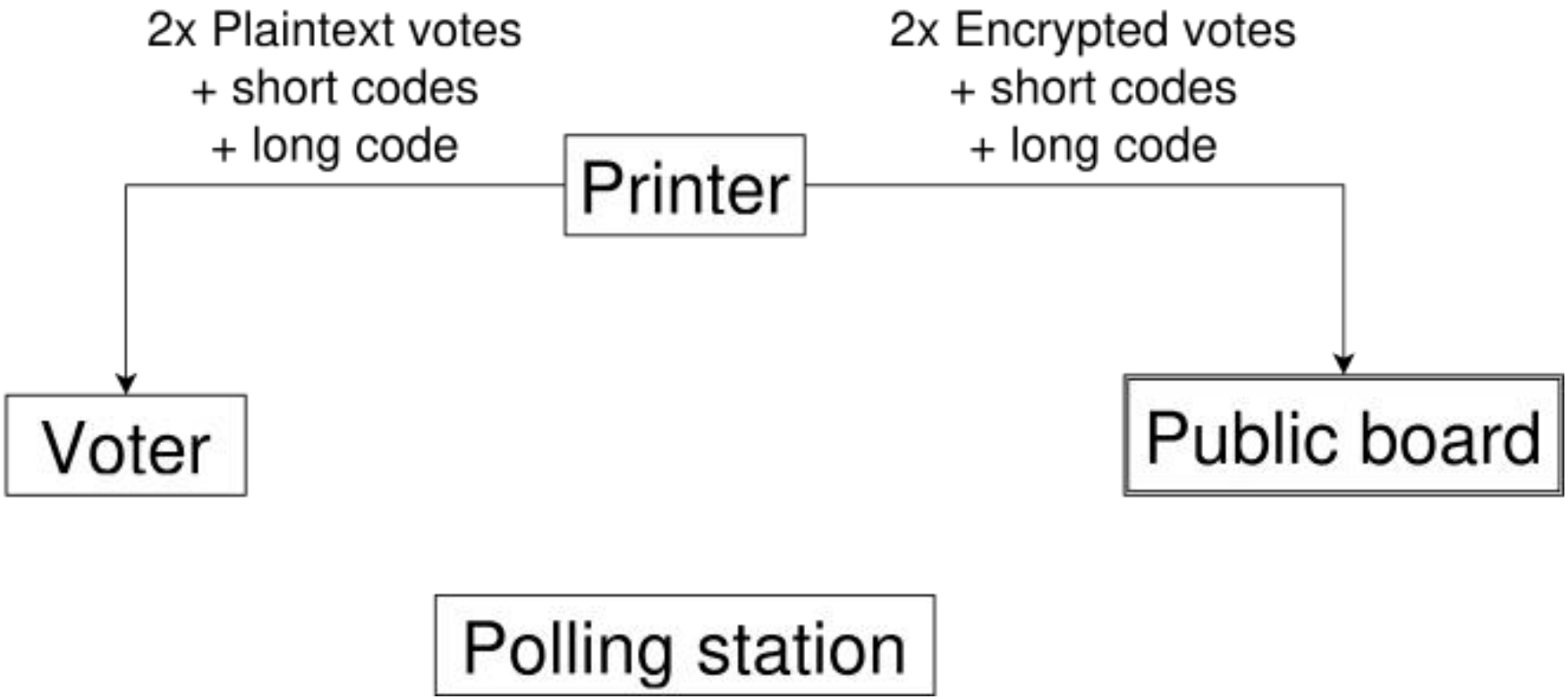


2022 Conservative Party leadership election

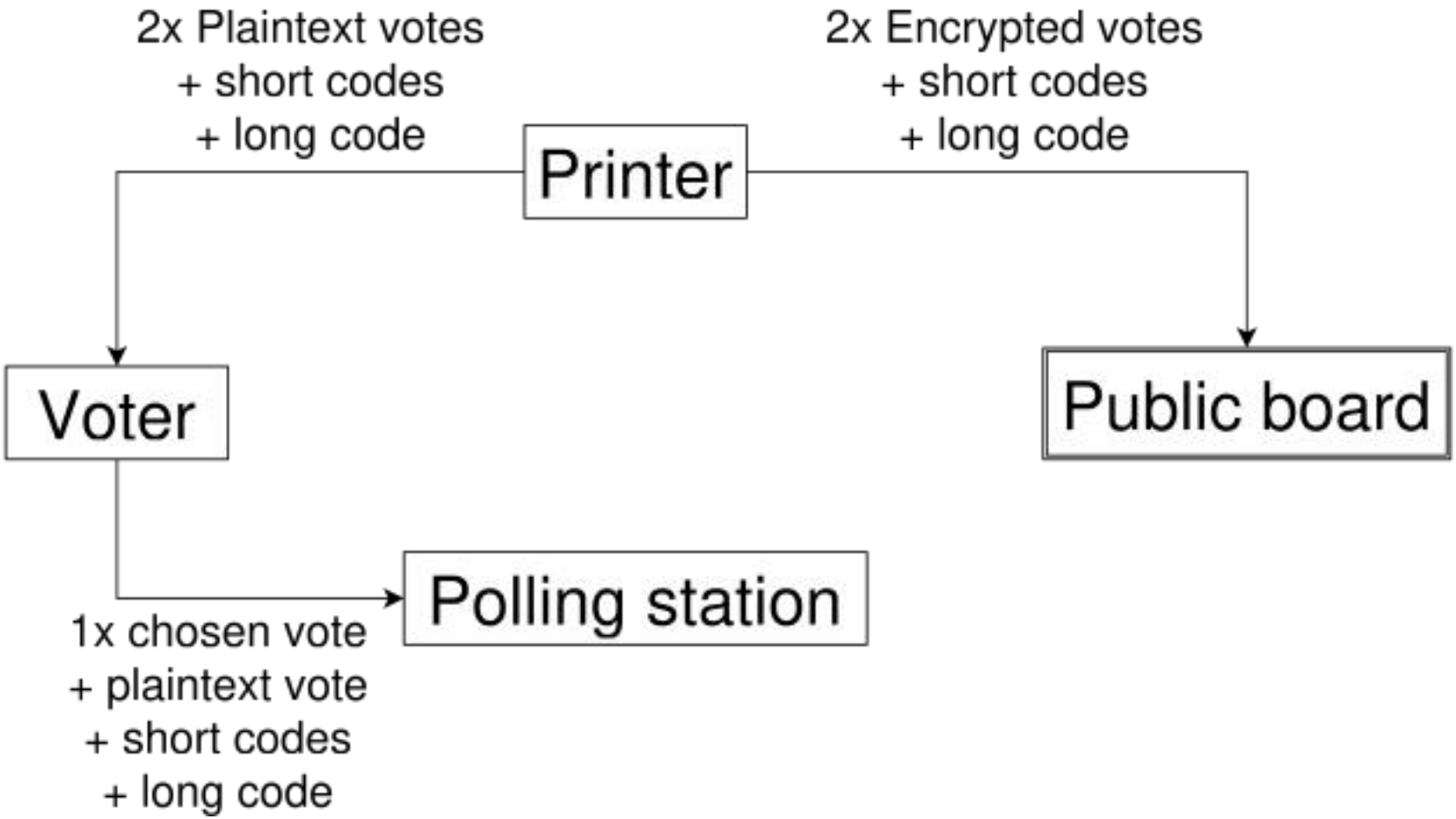
State of the art

		STROBE	RemoteVote	SAFE Vote	Vote-by-Mail	Vote&Check
privacy attacks	full privacy breach	😈	😈	😈	🛡️	🛡️
	targeted privacy breach	😈	😈	😈	🛡️	🛡️
	privacy breach by complaints	😈	😈	😈	😈	😈
verifiability attacks	vote flooding	😈	😈	🛡️	🛡️	🛡️
	clash attack	🛡️	😈	😈	😈	🛡️
	alter non verifying voters	😈	😈	😈	😈	😈
	weak eligibility	😈	😈	😈	😈	😈
	universal verifiability	🛡️	🛡️	🛡️	proxy	🛡️

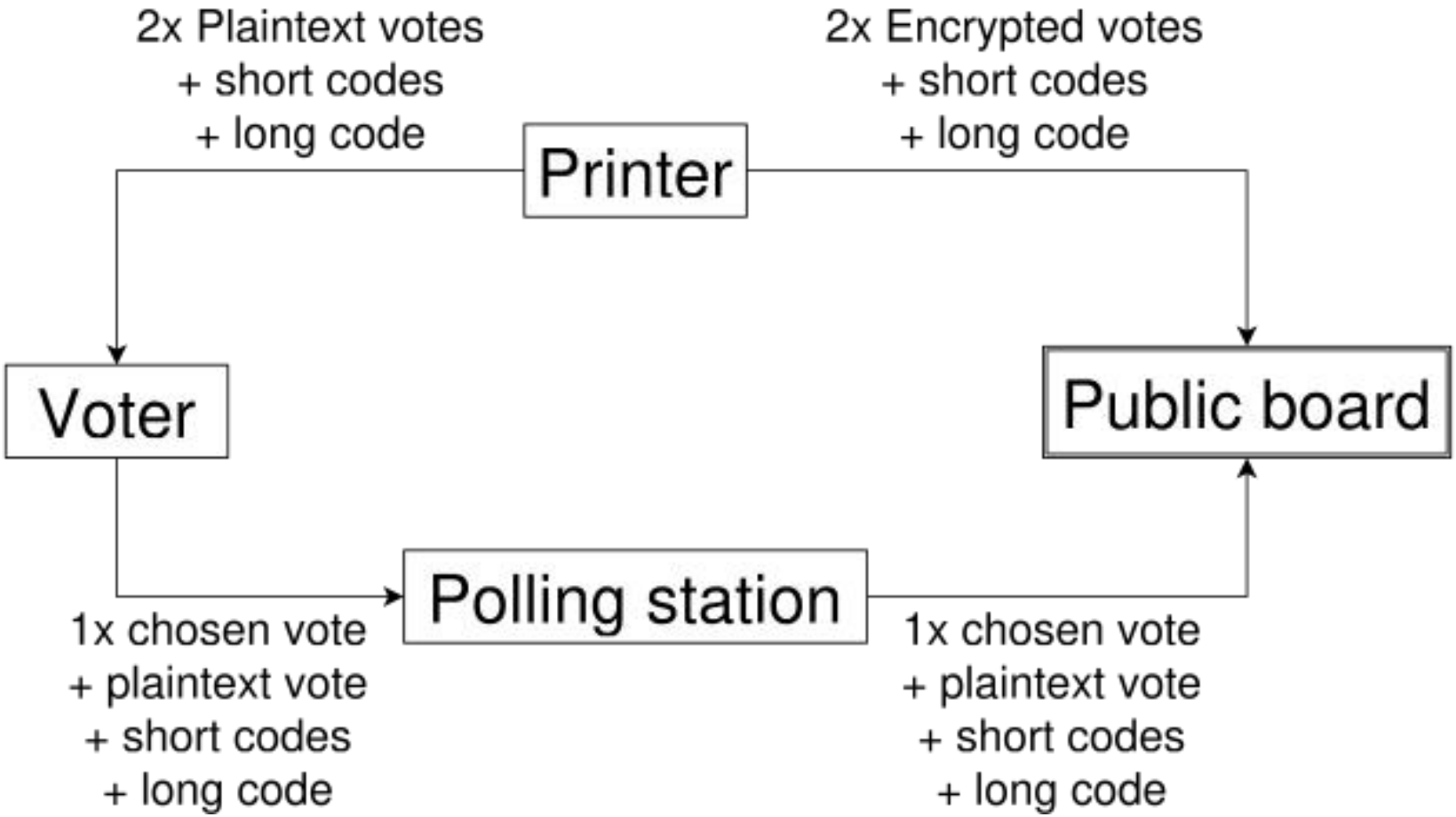
STROBE in a nutshell



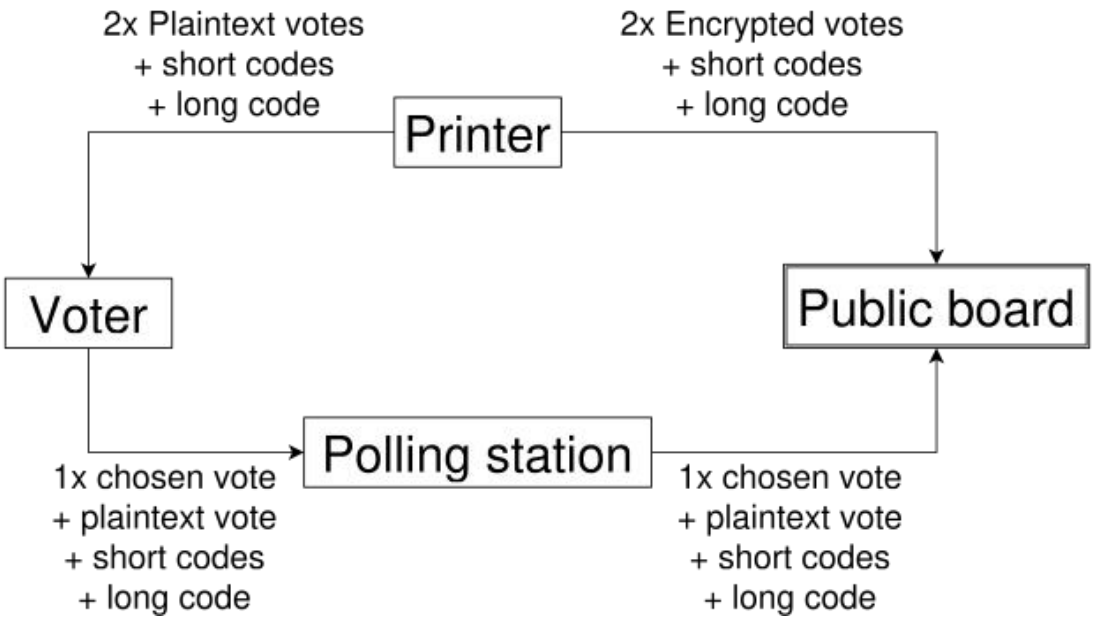
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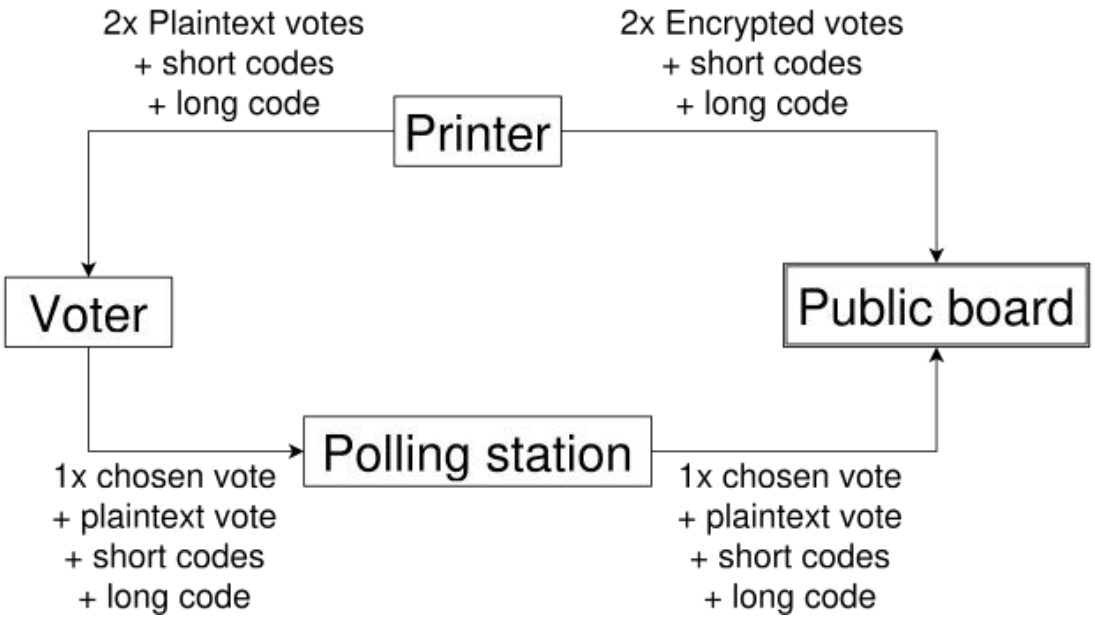


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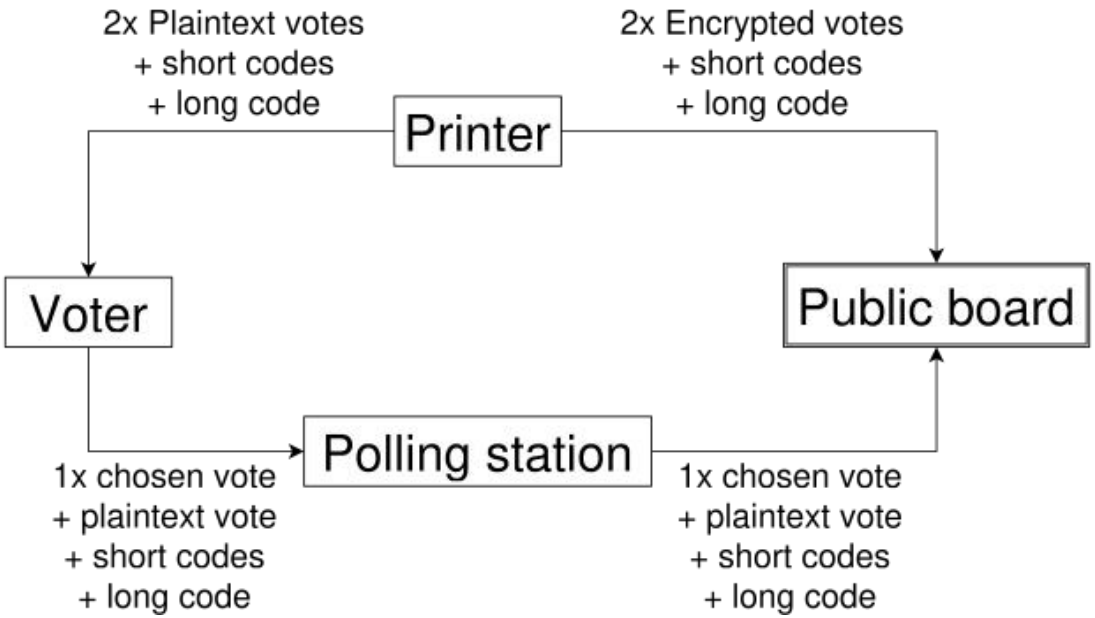
- Cast ballots are homomorphically decrypted
- Uncast ballot are spoiled

STROBE in a nutshell



Attack : full privacy breach !

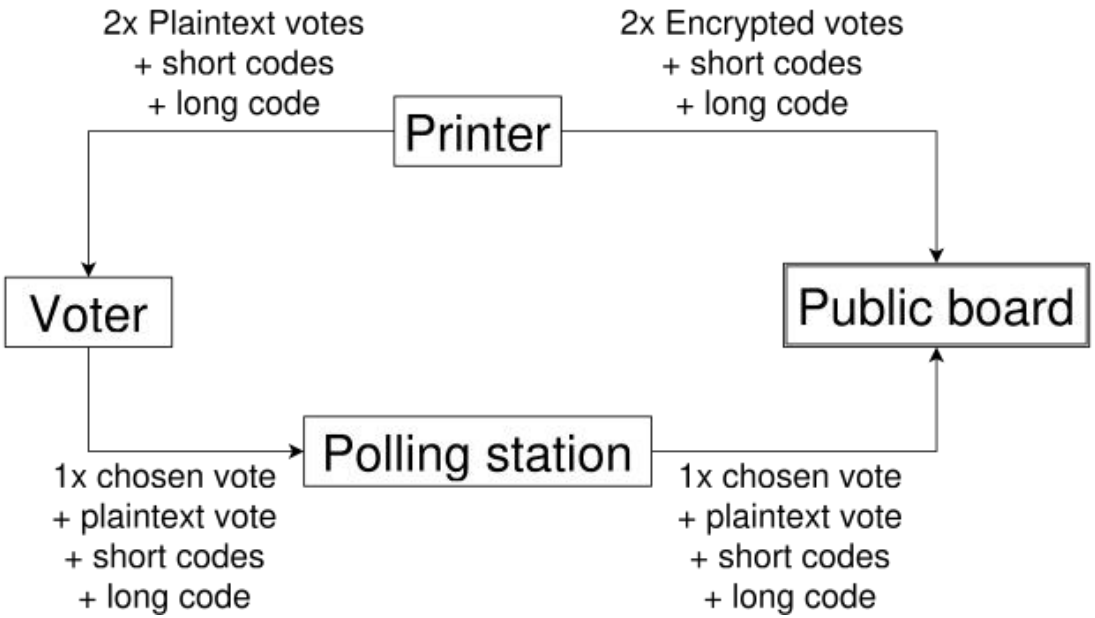
STROBE in a nutshell



Attack : full privacy breach !

Attack : targeted privacy breach !

STROBE in a nutshell



Attack : full privacy breach !

Attack : targeted privacy breach !

Attack : vote flooding !

Requierments

Goal: design a verifiable postal voting protocol with security proofs

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Constraints:

- Possible to vote "as usual"
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What does "good" mean?

Requierments

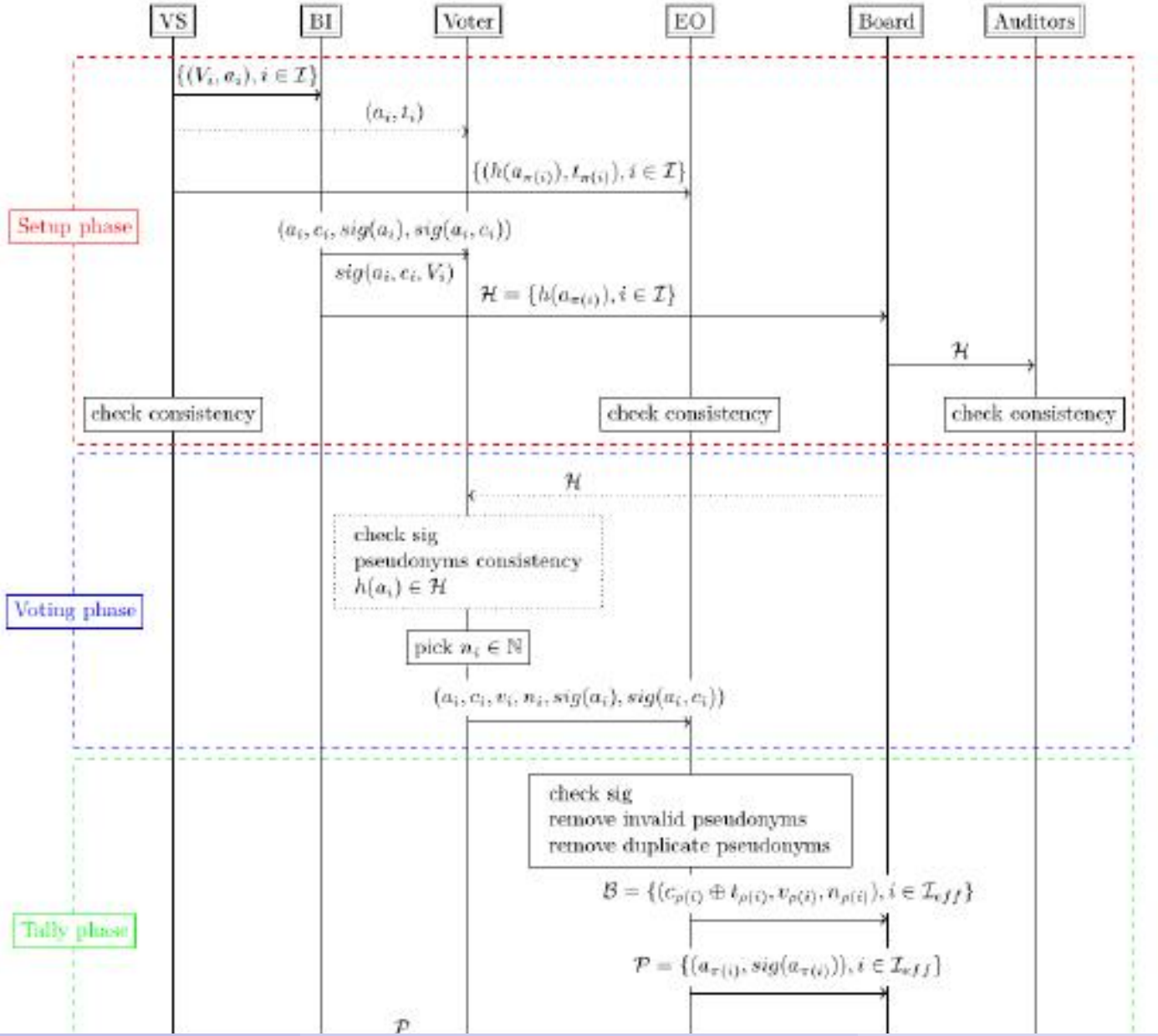
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Constraints:

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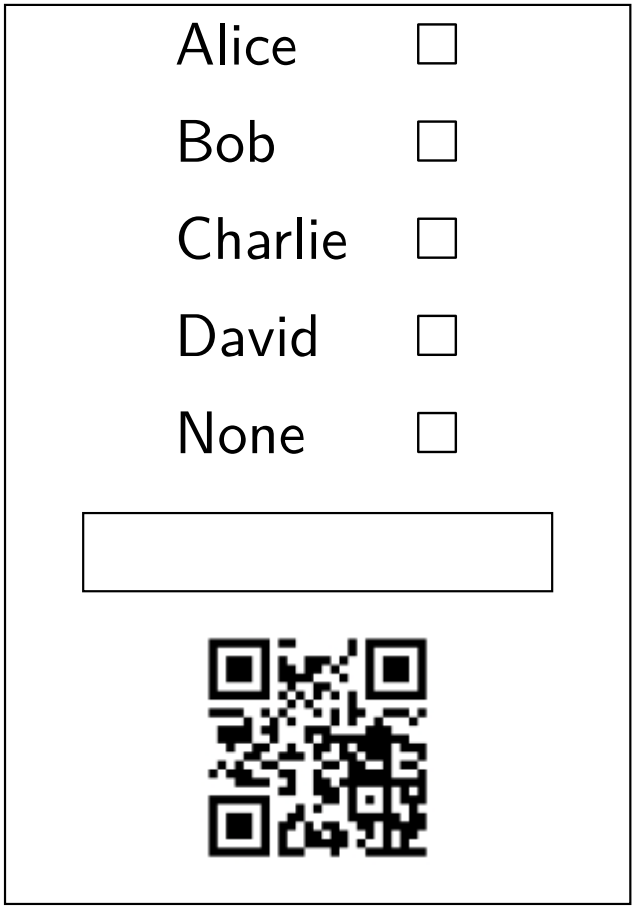
- vote secrecy
- integrity of the vote $\longrightarrow$ end-to-end verifiability

New proposal: Vote&Check



POV: you are the voter

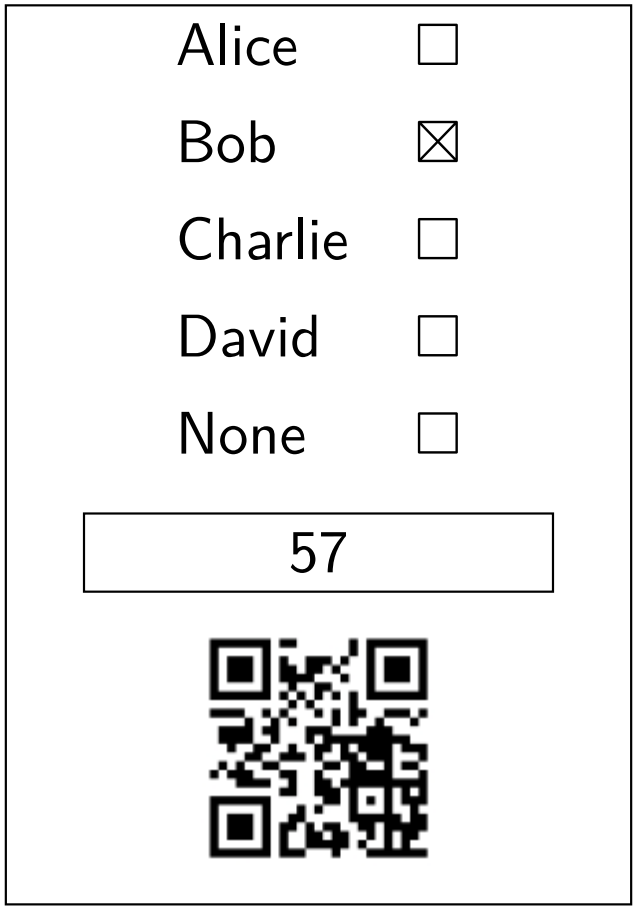
Step 1: setup





POV: you are the voter

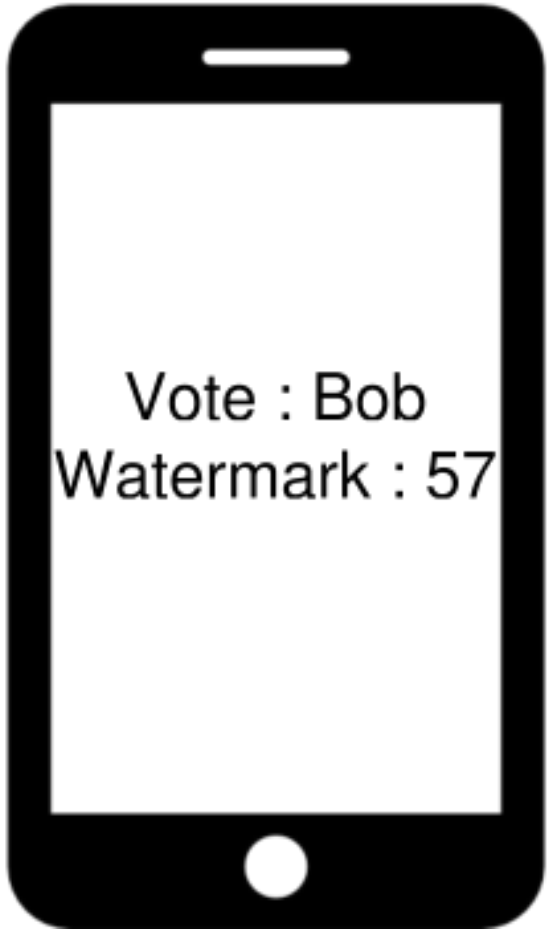
Step 2: voting



POV: you are the voter

Step 3: checking





Conception

Conception

Election Organiser

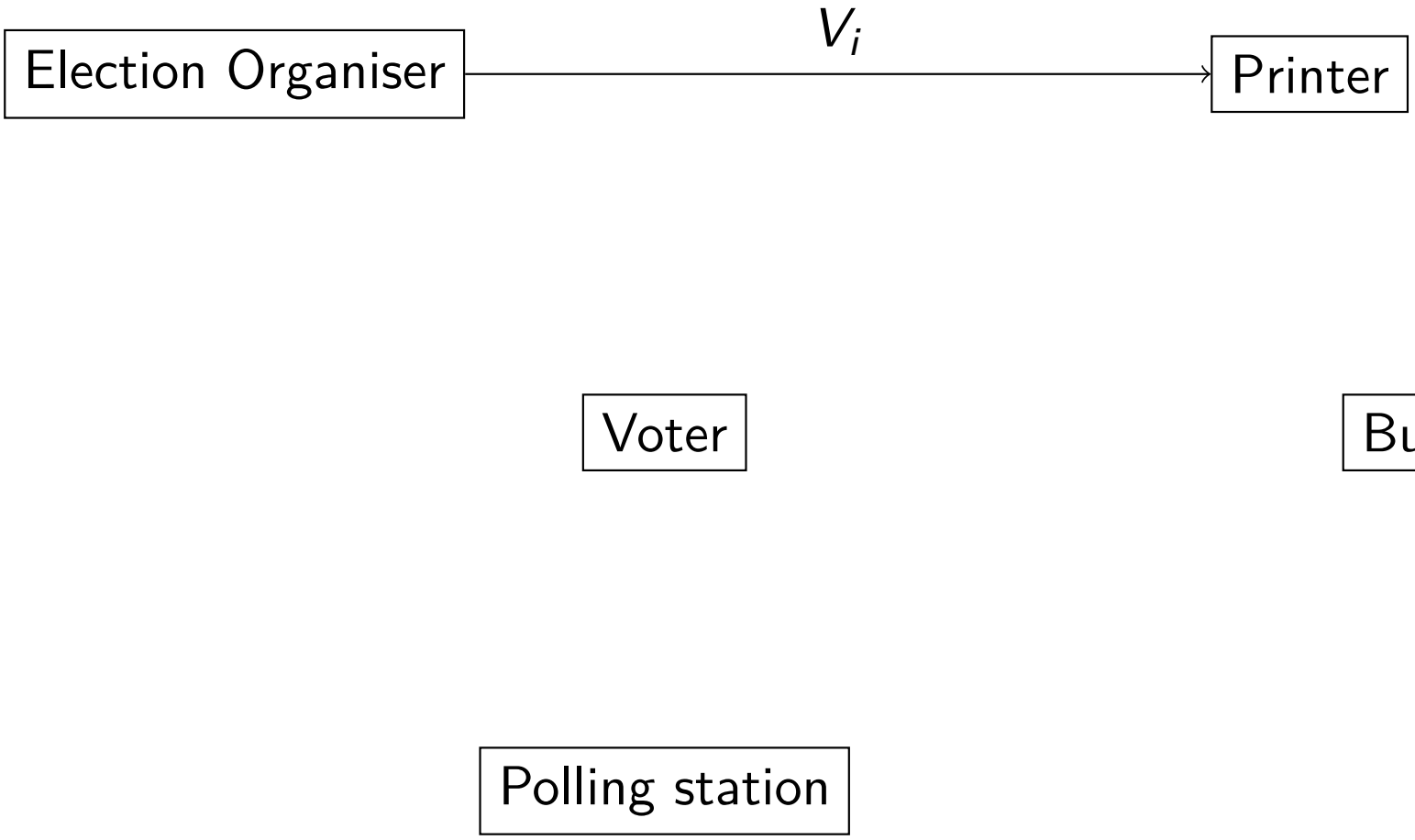
Printer

Voter

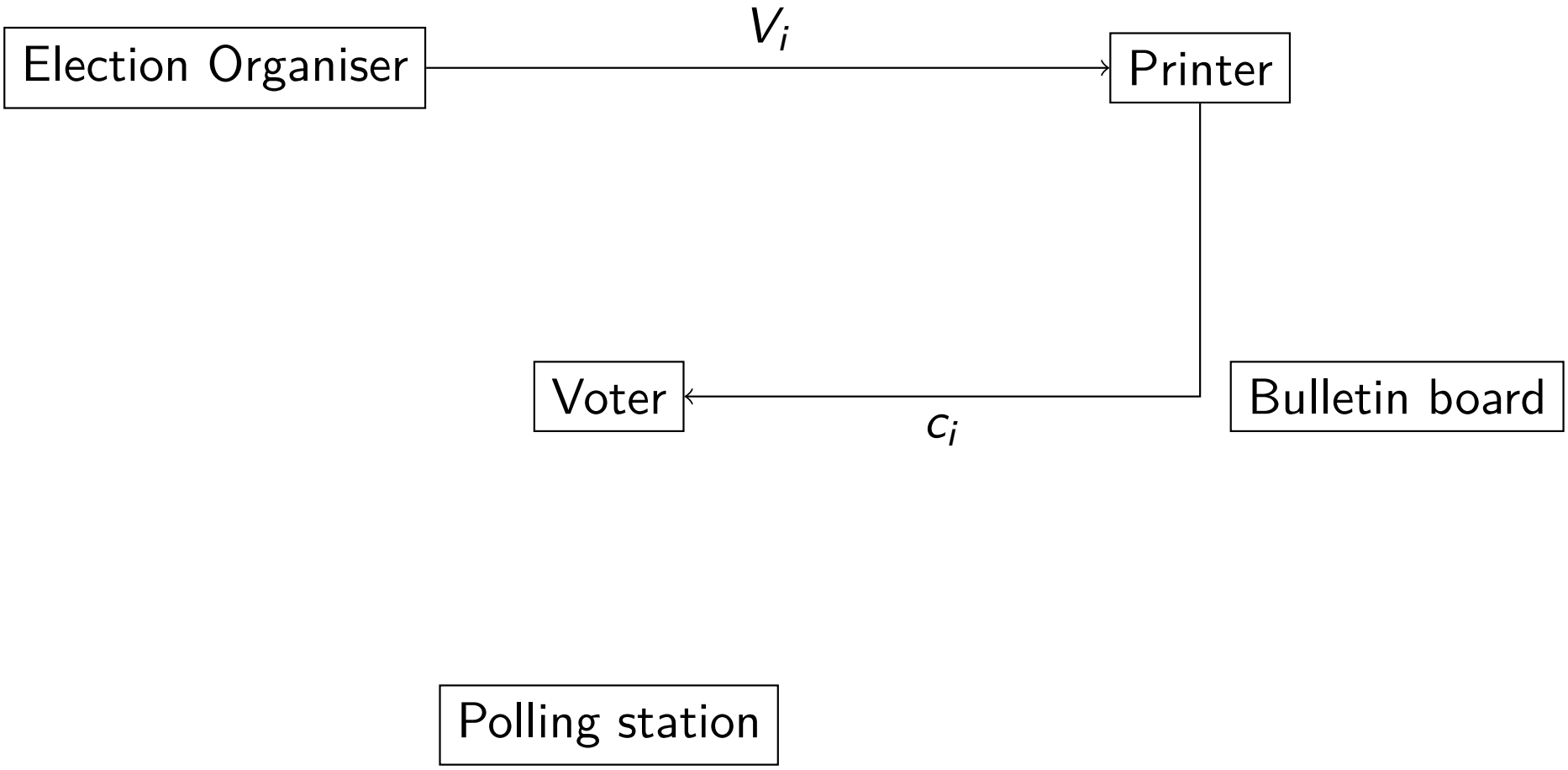
Bulletin board

Polling station

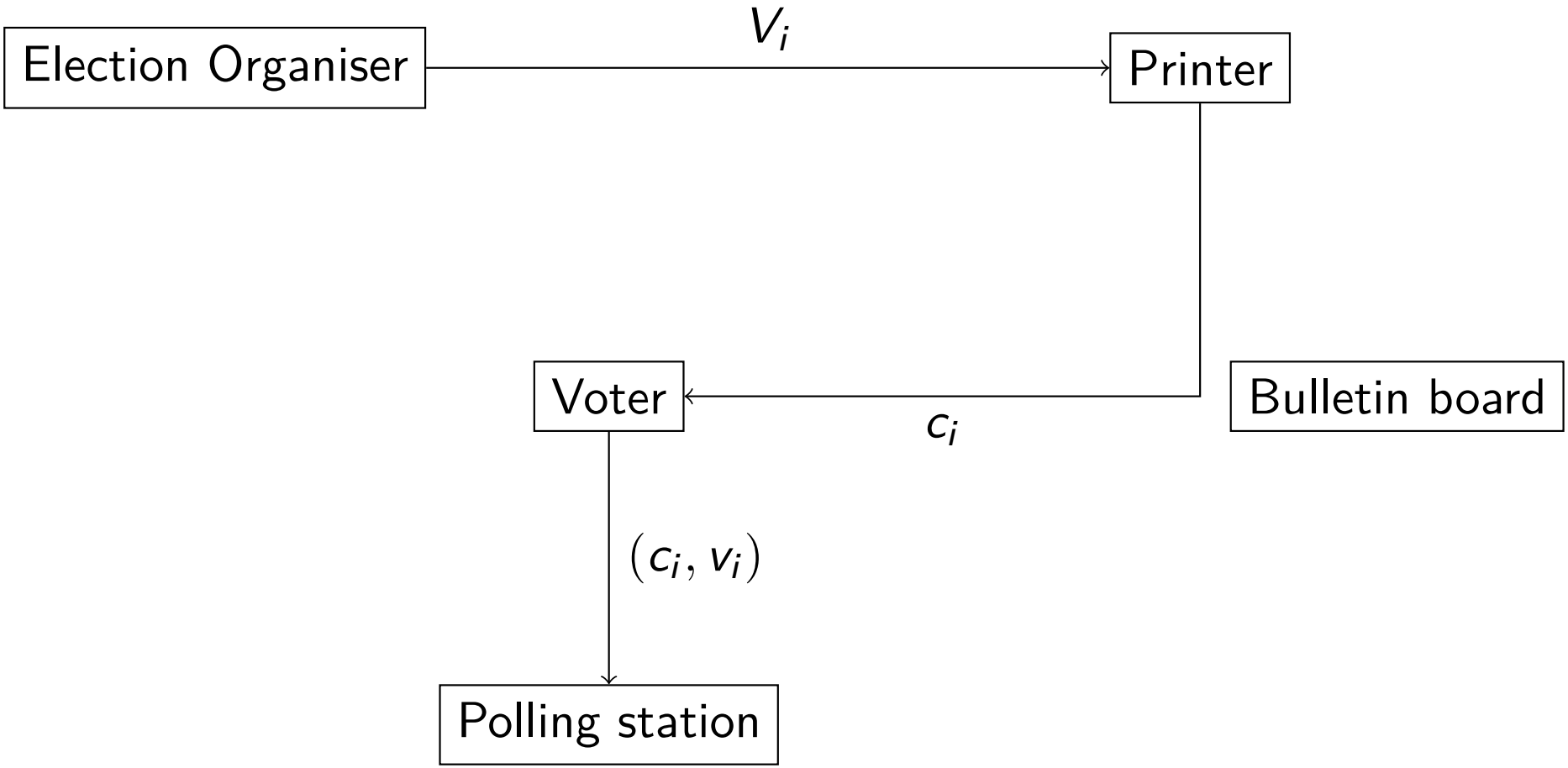
Conception


Election Organiser
V_i
Printer
Voter
Bulletin board
Polling station

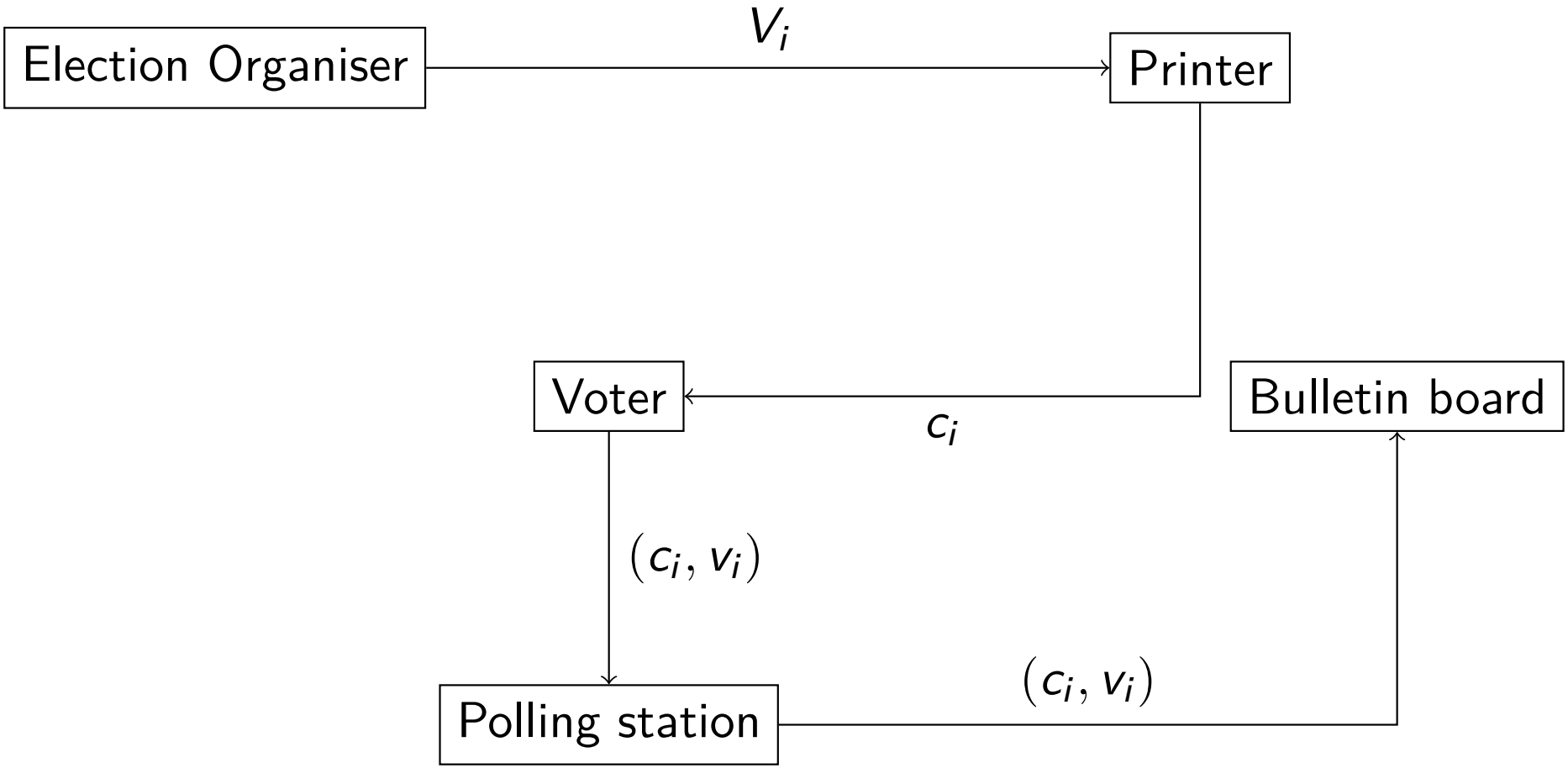
Conception


Election Organiser
V_i
Printer
Voter
c_i
Bulletin board
Polling station

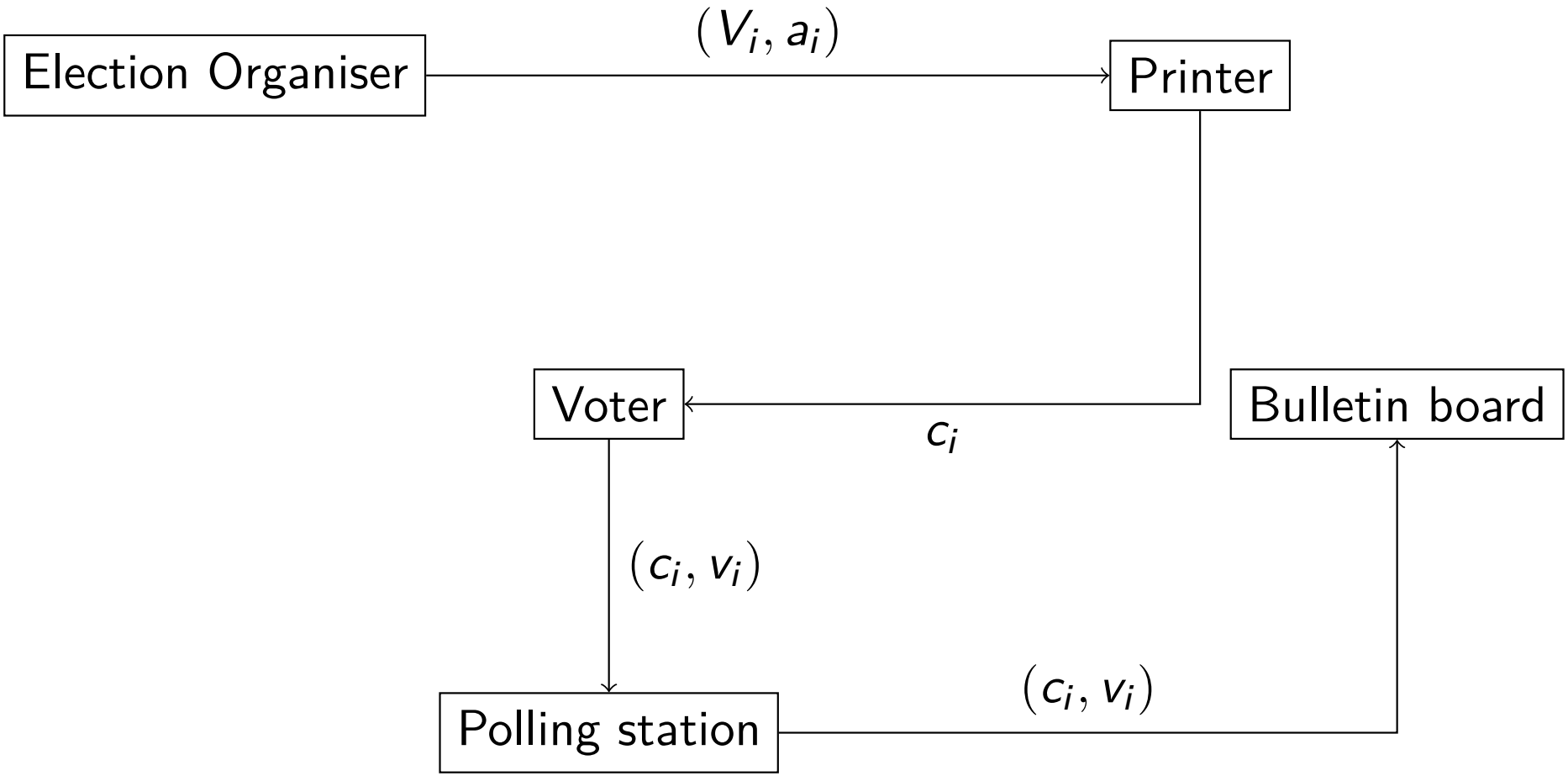
Conception


Election Organiser
V_i
Printer
Voter
Bulletin board
c_i
(c_i, v_i)
Polling station

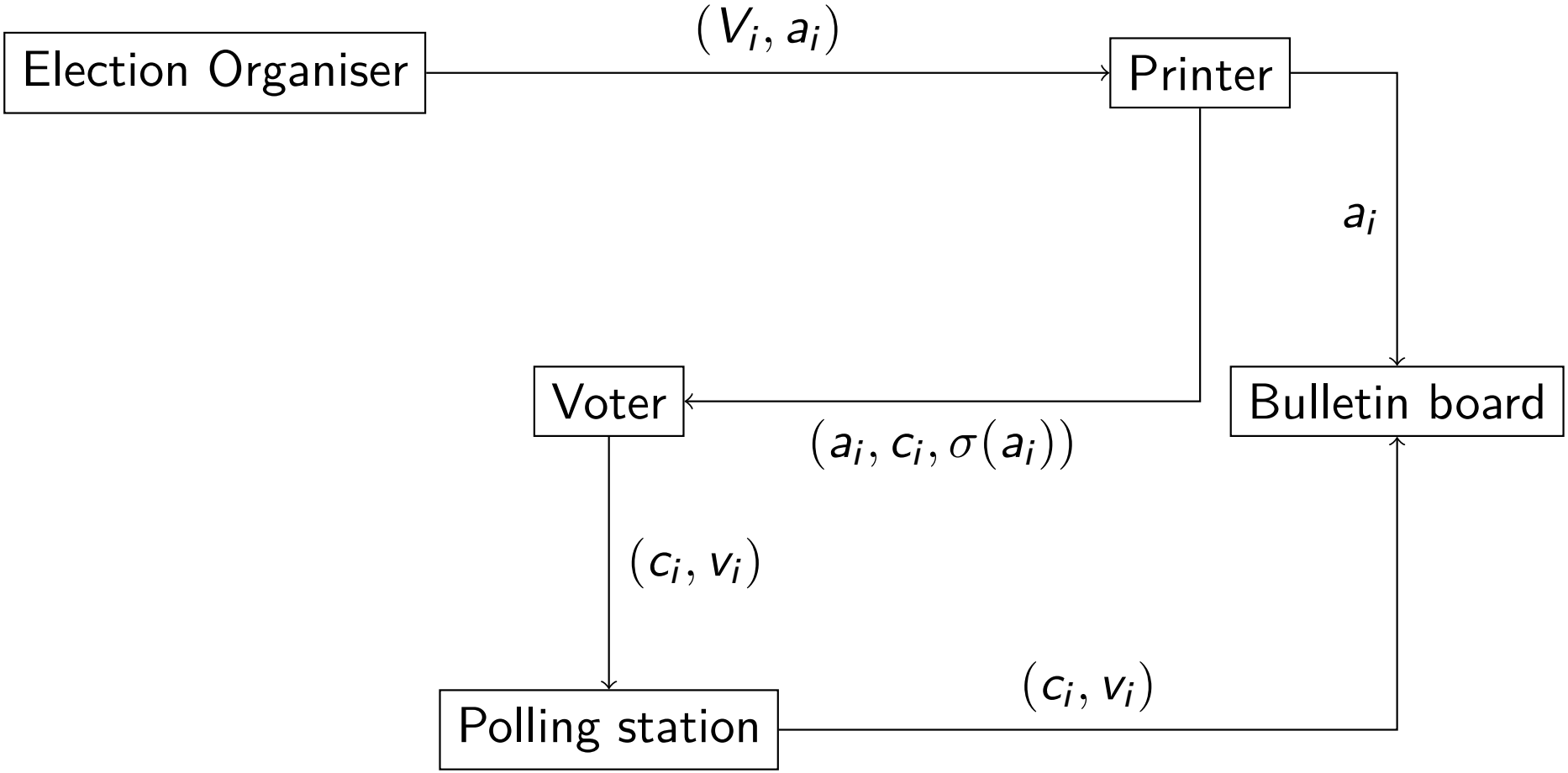
Conception


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Polling station
(c_i, v_i)

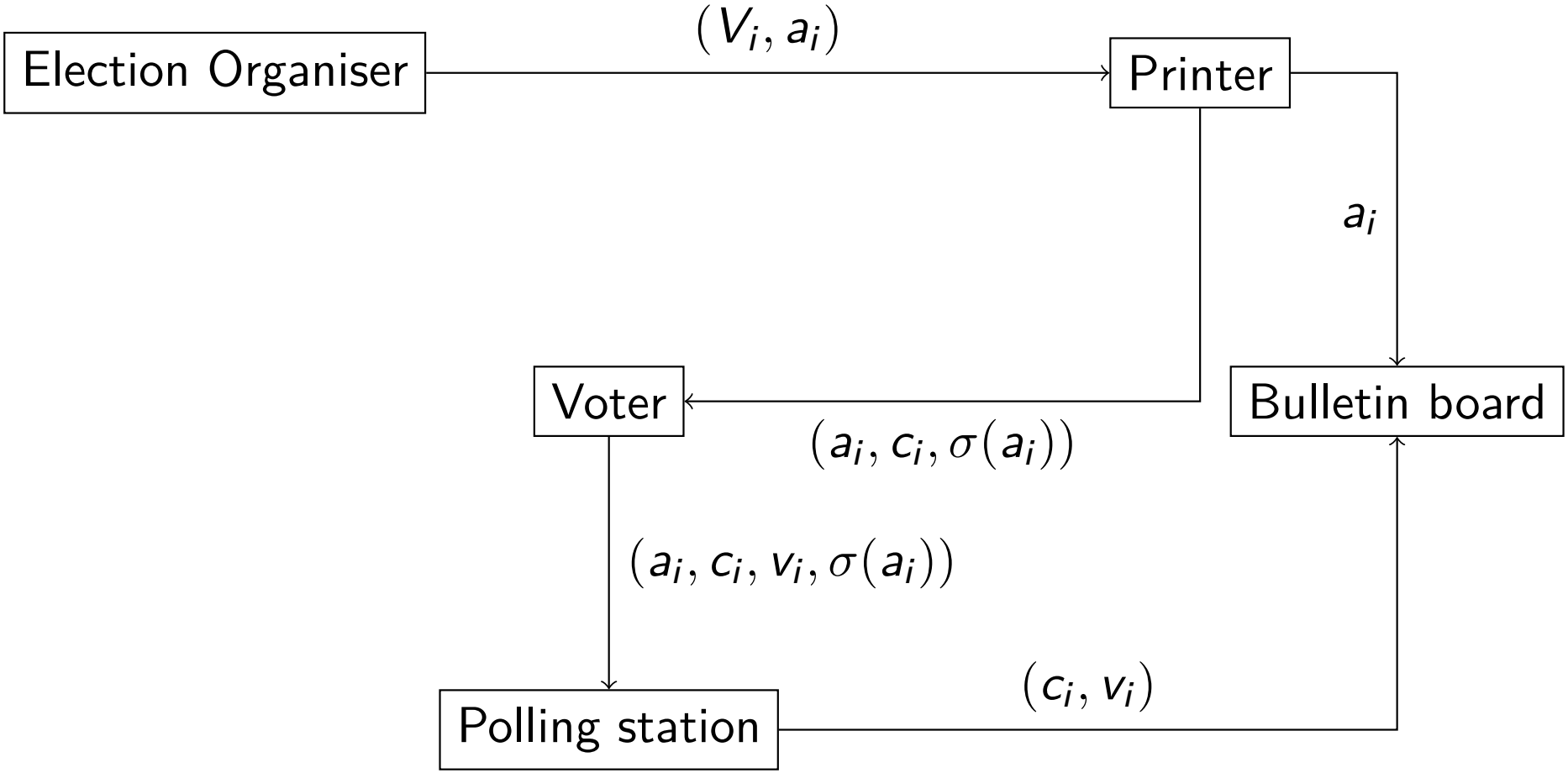
Conception


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(V_i, a_i)
Printer
Voter
c_i
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(c_i, v_i)
Polling station
(c_i, v_i)

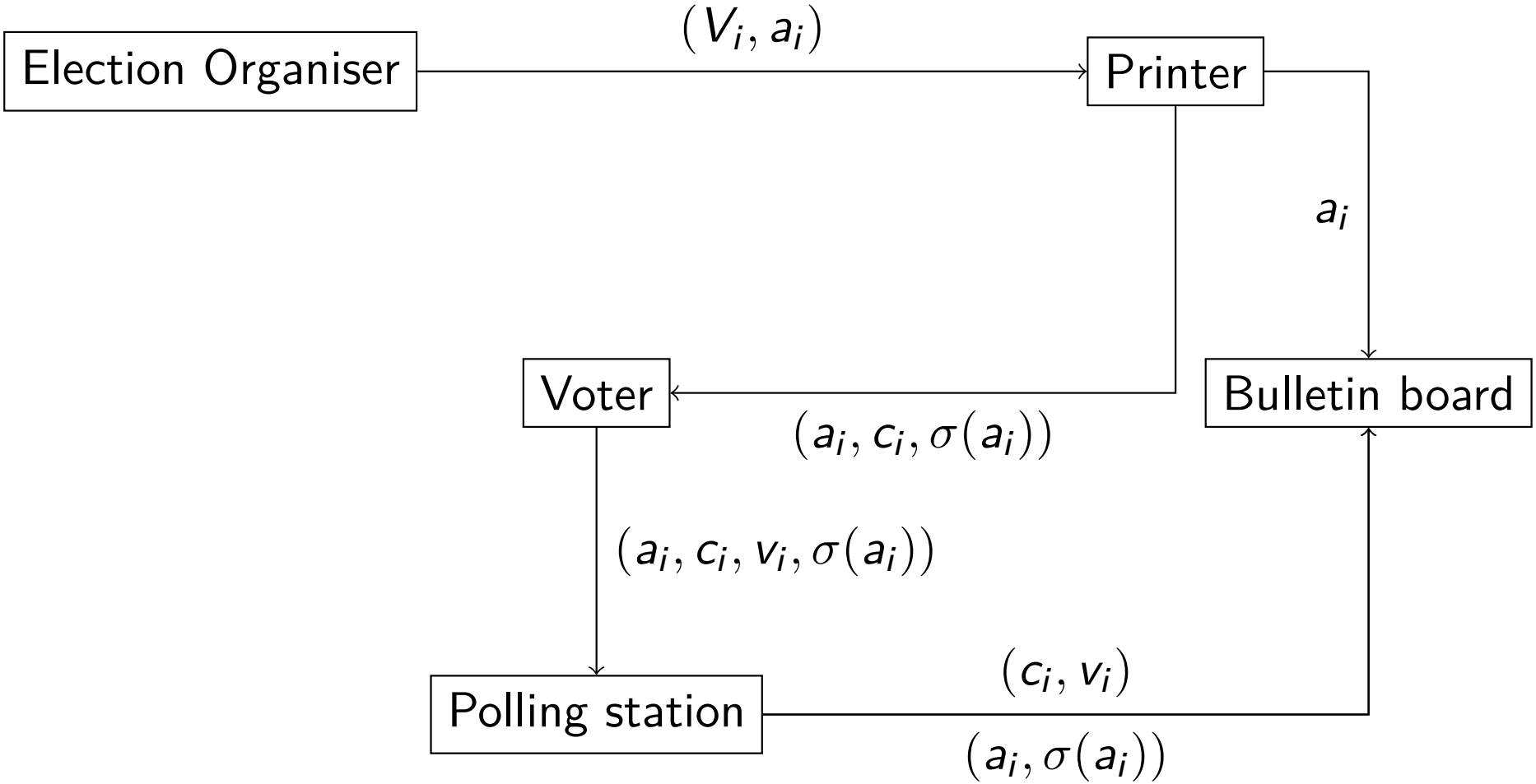
Conception


Election Organiser
(V_i, a_i)
Printer
a_i
Voter
Bulletin board
$(a_i, c_i, \sigma(a_i))$
(c_i, v_i)
(c_i, v_i)
Polling station

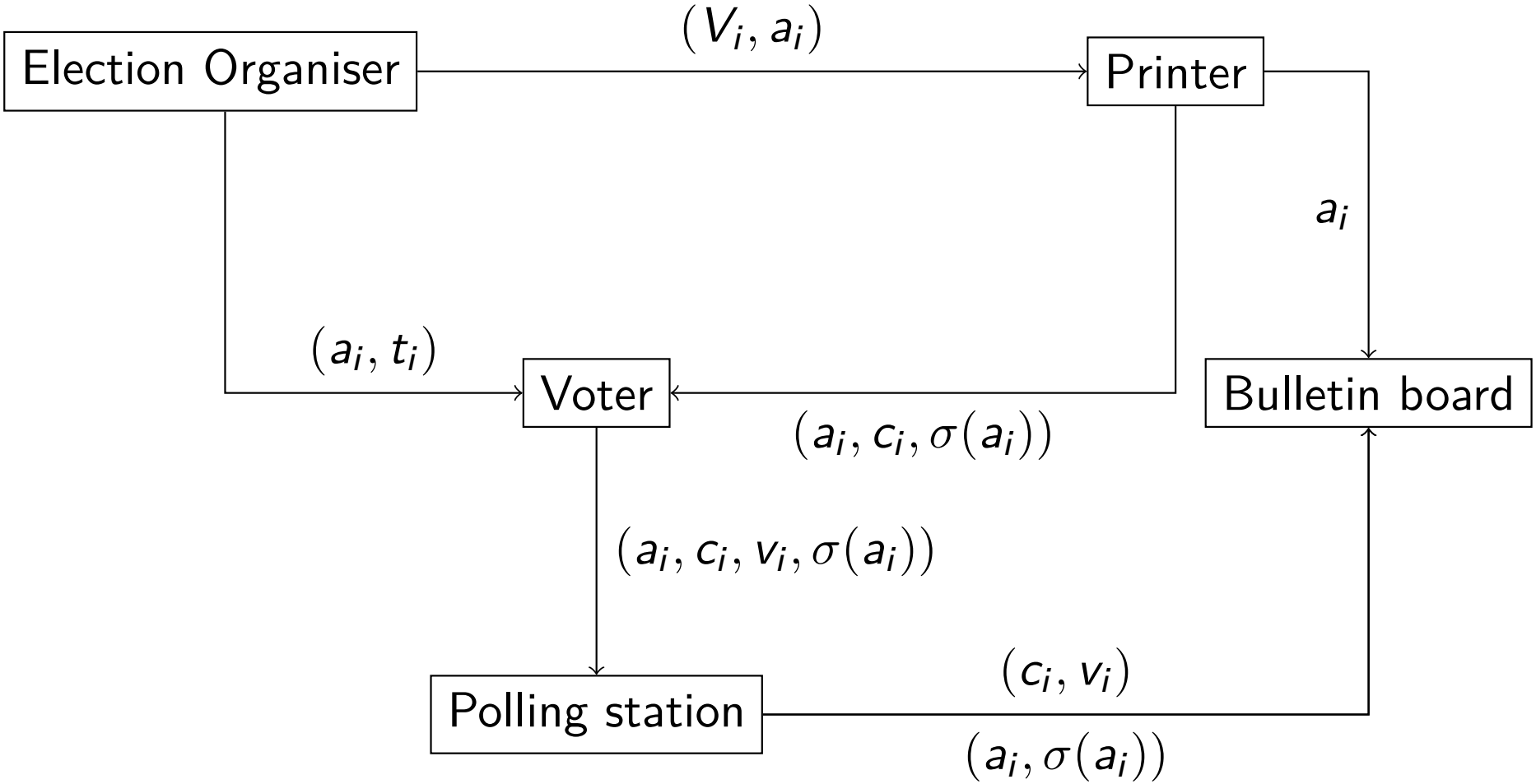
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(c_i, v_i)
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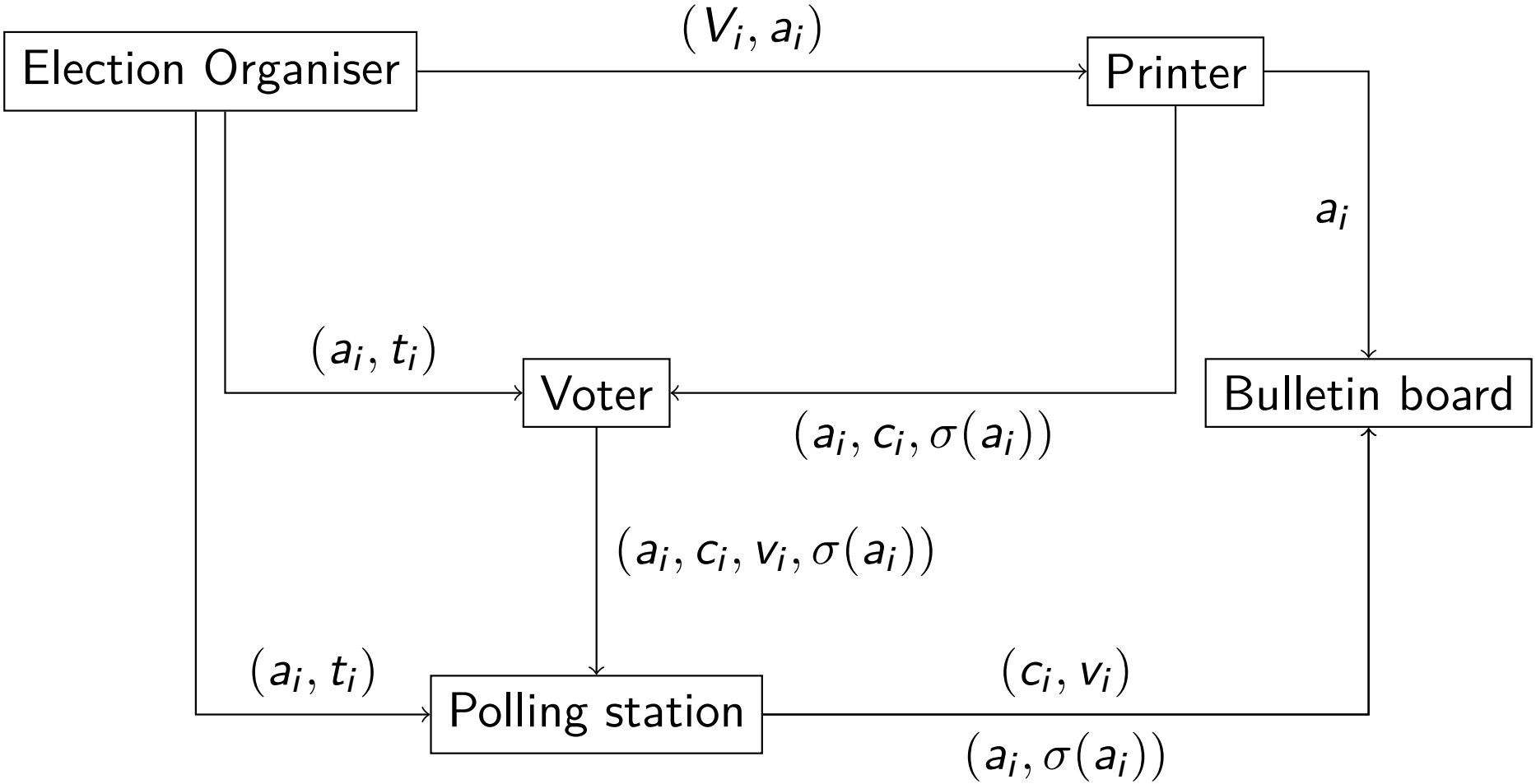
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$(a_i, c_i, \sigma(a_i))$
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(c_i, v_i)
$(a_i, \sigma(a_i))$

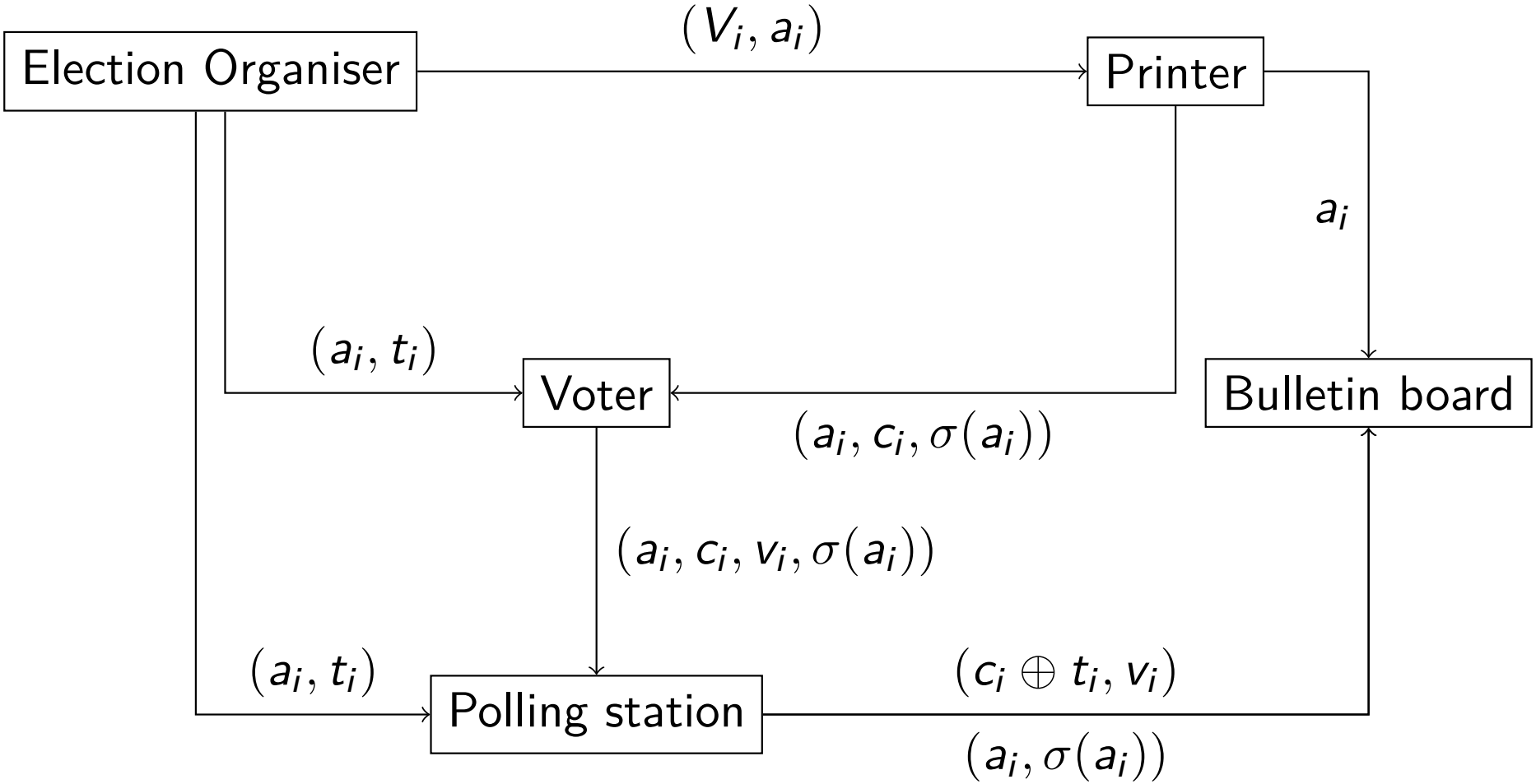
Conception


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Printer
a_i
(a_i, t_i)
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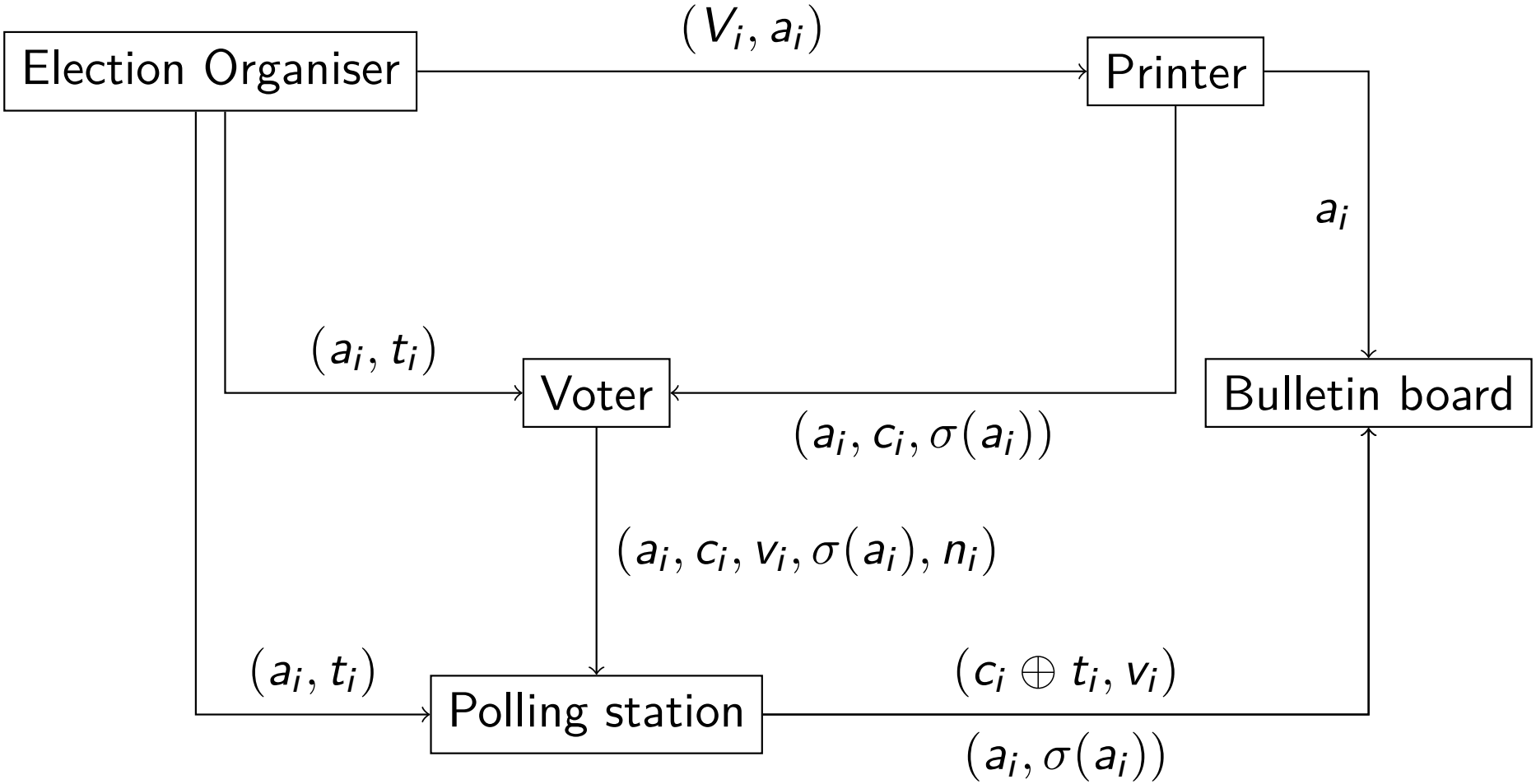
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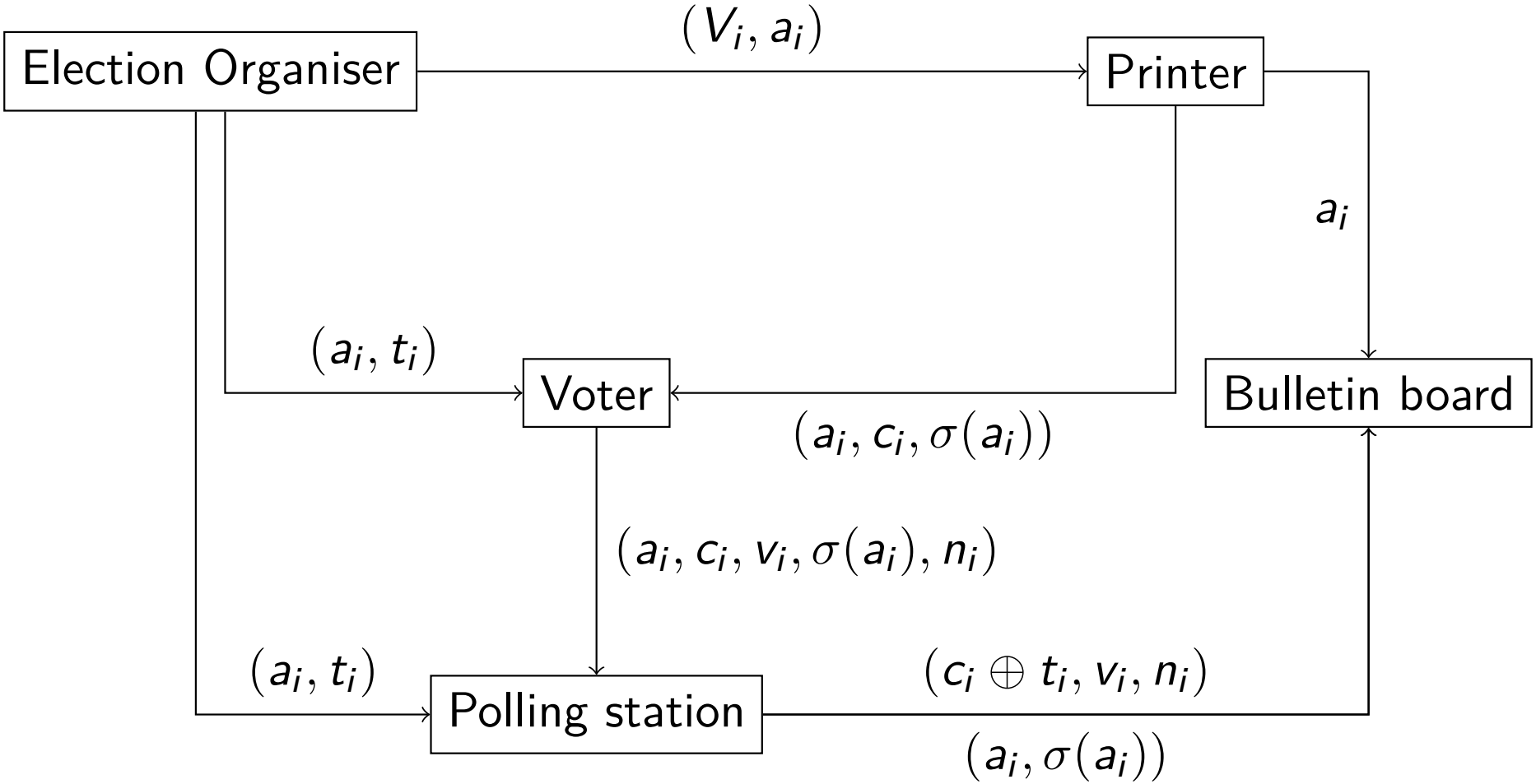
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$(a_i, c_i, v_i, \sigma(a_i))$
(a_i, t_i)
Polling station
$(c_i \oplus t_i, v_i)$
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Security properties

			Privacy	E2E Verifiability	Weak verifiability	Individual verifiability
Tracker Server	Printer	Election Office				
H	H	H	✓	✓	✓	✓
C	H	H	✓	✓	✓	✓
H	C	H	✓	✗	✗	✓
H	H	C	✓	✗	✓	✓
H	C	C	✗	✗	✗	✓
C	H	C	✗	✗	✓	✓
C	C	H	✗	✗	✗	✓
C	C	C	✗	✗	✗	✓

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H	C	H	✓	✗	✗	✓
H	H	C	✓	✗	✓	✓
H	C	C	✗	✗	✗	✓
C	H	C	✗	✗	✓	✓
C	C	H	✗	✗	✗	✓
C	C	C	✗	✗	✗	✓

Weak verification:

- If you verify, your vote is counted
- If you do not, your vote may be droped *or changed*
- Adversary cannot add ballots (except corrupt voters)

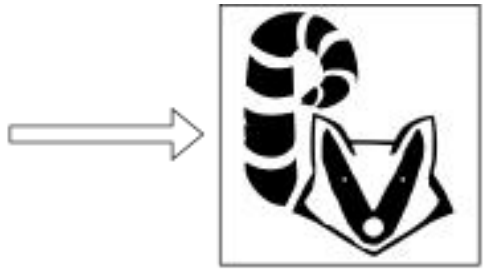
Verification



- Formal proof written with the Proverif tool
- Verifiability required a dedicated framework

The Proverif tool

```
in(mailbox, (ballot, signature));
if check(signature, ballot, pk_Printer) then
out(ballot_box, (ballot, v));
event Voted(v).
```



```
true
false
cannot be proved
timeout
```

```
query event(Counted(v)) ==>
      event(Voted(v)).
```

Results from formal analysis

- New properties: weak verification and privacy without complaint
- More precise specification
- Improving the framework

Conclusion

- Original attacks on existing protocols
- New protocol proposed
- Formal security proofs

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Future work

- Finish the proofs
- Write the paper
- ...